How-To Library

Informatica

Address Verification 6.x Best Practices for the United States

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Abstract

This document tells you how to connect the elements in a United States address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in the United States.

Supported Versions

- Informatica Address Verification (Cloud) 6.x
- Informatica Address Verification (On-Premises) 6.x

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Introduction

Informatica Address Verification can validate and enhance addresses in the United States. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate JSON elements for United States address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to analyze the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for United States addresses:

- BATCH
- CERTIFIED
- GEOCODE TO ADDRESS

- INTERACTIVE
- QUICK CAPTURE

Character Set

United States reference address databases store address data in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The United States Postal Service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: HOUSE NUMBER STREET NAME [SUB-BUILDING INFORMATION]
Line 3: CITY STATE ZIP CODE
Line 4: COUNTRY
```

The following sample address shows the structure:

```
2101 MASSACHUSETTS AVE NW
WASHINGTON DC 20008-2811
USA
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if two or more of the following items are absent from the address:

- AdministrativeDivision
- Locality
- ZIP Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street	MASSACHUSETTS AVE NW	Street Name, Street PostDescriptor, Street PostDirectional
House Number	2101	HouseNumber
City	WASHINGTON	Locality
State	DC	AdministrativeDivision
ZIP Code	20008-2811	PostalCode
Country	USA	Country

Input and Result JSON in Address Verification (On-Premises)

The following example shows how you can map the address information to the JSON address elements in Address Verification (On-Premises).

You enter the following input address:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {}
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": { "Country": "USA" },
        "PreformattedData": {
          "PostalFormattedAddressLines": [
            "2101 MASSACHUSETTS AVE NW", "WASHINGTON DC 20008-2811"
          ]
        }
      }
    ]
}
}
```

Informatica Address Verification (On-Premises) returns the following result:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {}
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": { "Country": "USA" },
        "PreformattedData": {
          "PostalFormattedAddressLines": [
            "2101 MASSACHUSETTS AVE NW", "WASHINGTON DC 20008-2811"
          ]
        }
      }
    ],
    "Outputs": [
      {
        "ResultInfo": {
          "ProcessStatus": "V",
          "ProcessModeUsed": "Batch",
          "DeliverAdditionalSuggestions": "No",
          "ResultCount": 1,
          "ResultCountOverflow": false,
          "ResultCountries": [ "USA" ]
        },
        "Results": [
          {
            "Variants": [
              {
                "StatusValues": {
                  "AddressType": "S",
                  "LanguageISO3": "ENG",
                  "UsedVerificationLevel": "All",
                  "MatchPercentage": "100.00",
                  "Script": "Native Primary",
                  "AddressCount": "",
                  "ResultQuality": 6
                },
                "AddressElements": {
                  "Street": [ { "Value": "MASSACHUSETTS AVE NW" } ],
```

```
                    "HouseNumber": [ { "Value": "2101" } ],
                    "Locality": [ { "Value": "WASHINGTON" } ],
                    "PostalCode": [ { "Value": "20008-2811" } ],
                    "AdministrativeDivision": [
                      { "Value": "DC" }, { "Value": "DISTRICT OF COLUMBIA" }
                    ],
                    "Country": [ { "Code": "USA", "Name": "UNITED STATES" } ]
                  },
                  "PreformattedData": {
                    "PostalDeliveryAddressLines": [
                      { "Value": "2101 MASSACHUSETTS AVE NW" }
                    ],
                    "PostalFormattedAddressLines": [
                      { "Value": "2101 MASSACHUSETTS AVE NW" }, { "Value": "WASHINGTON DC
20008-2811" }
                    ],
                    "PostalLocalityLine": { "Value": "WASHINGTON DC 20008-2811" }
                  }
                }
              ]
            }
          ]
        }
      ]
    }
}
```

Input and Result JSON in Address Verification

The following example shows how you can verify a United States address in a REST call to Address Verification.

You submit the following request:

```
POST /AV6/v1/process HTTP/1.1
Host: us-av.informaticadaas.com
Content-Type: application/json
{
 "Login": "***",
 "Password": "***",
 "CampaignId": "Campaign111",
 "UseTransactions": "TEST",
 "Request": {
  "Parameters": {
   "Mode": "Batch",
   "CountryDetermination": {},
   "DataSetDetermination": {},
   "CountrySets": [{
     "Process": {
      "AlternativeHandling": {},
      "VerificationLevel": {
       "PreferredVerificationLevel": "All",
       "MinimumVerificationLevel": "All"
      },
      "Geocode": {
       "SearchRadius": "50"
      },
      "Enrichments": {
       "Geocoding": {
        "Rooftop": false,
        "ArrivalPoint": false,
        "StreetCenter": false,
        "PostalCodeCenter": false,
        "LocalityCenter": false
       },
       "Global": false,
       "CountrySpecific": false,
       "CAMEO": false
      }
     },
     "Input": {
```

```
        "SingleAddressLineDataType": "UndelimitedSingleLine",
        "SingleAddressLineDelimiter": "Comma"
       },
       "Result": {
        "MaxResultCount": 100,
        "NumericRangeExpansion": {
         "RangesToExpand": "None",
         "RangeExpansionType": "Flexible"
        }
       },
       "OutputDetail": {
        "PreformattedData": {
         "PostalFormattedAddressLines": true,
         "PostalDeliveryAddressLines": true,
         "PostalRecipientLines": true,
         "PostalLocalityLine": true,
         "StreetWithNumber": false,
         "NumberAndSubBuilding": false,
         "SingleAddressLine": true,
         "SingleAddressLineDelimiter": "Space"
        },
        "SubItems": true,
        "ElementStatus": true,
        "OutputStatusForEmptyElements": true
       },
       "Standardizations": [{
         "Default": {
          "PreferredScript": {
           "Script": "NativePrimary",
           "TransliterationType": "Default",
           "LimitLatinCharacters": "NoLimit"
          },
          "FormatWithCountry": false,
          "CountryNameLanguage": "ENG",
          "CountryCodeType": "ISO3",
          "MaxItemLength": 255,
          "Casing": "PostalAdmin",
          "DescriptorLength": "Database",
          "AliasHandling": "PostalAdmin"
         }
        }
       ]
      }
     ]
    },
    "IO": {
     "Inputs": [{
       "PreformattedData": {
        "PostalFormattedAddressLines": [
         "2101 MASSACHUSETTS AVE NW ", "WASHINGTON DC 20008-2811 ", "USA "
        ]
       }
      }
     ]
    }
   }
  }
```

Informatica Address Verification returns the following response:

```
{
   "Status": "Ok",
   "StatusDescription": "No error",
   "TransactionInfo":    {
      "TotalTransactionsCharged": 1,
      "TransactionPools": [      {
         "Type": "BATCH",
         "TransactionsCharged": 1
      }]
   },
   "CampaignId": "Campaign111",
```

```
"Response": [   {
   "ResultInfo":       {
      "ProcessStatus": "V",
      "ProcessModeUsed": "Batch",
      "ResultCount": 1,
      "ResultCountOverflow": false,
      "ResultCountries": ["USA"]
   },
   "Results": [{"Variants": [      {
      "StatusValues":          {
         "AddressType": "S",
         "LanguageISO3": "ENG",
         "UsedVerificationLevel": "None",
         "MatchPercentage": "100.00",
         "Script": "Database",
         "AddressCount": "",
         "ResultQuality": 6
      },
      "AddressElements":          {
         "Street": [            {
            "Value": "MASSACHUSETTS AVE NW",
            "SubItems":               {
               "Name": "MASSACHUSETTS",
               "PostDescriptor": "AVE",
               "PostDirectional": "NW"
            },
            "Status":               {
               "MatchStatus": "8",
               "ResultStatus": "1",
               "PostalRelevance": "2"
            }
         }],
         "HouseNumber": [            {
            "Value": "2101",
            "SubItems": {"Number": "2101"},
            "Status":               {
               "MatchStatus": "6",
               "ResultStatus": "1",
               "PostalRelevance": "2"
            }
         }],
         "Locality": [            {
            "Value": "WASHINGTON",
            "SubItems": {"Name": "WASHINGTON"},
            "Status":               {
               "MatchStatus": "8",
               "ResultStatus": "1",
               "PostalRelevance": "2"
            }
         }],
         "PostalCode": [            {
            "Value": "20008-2811",
            "SubItems":               {
               "Base": "20008",
               "AddOn": "2811"
            },
            "Status":               {
               "MatchStatus": "8",
               "ResultStatus": "1",
               "PostalRelevance": "2"
            }
         }],
         "AdministrativeDivision":            [
                        {
            "Value": "DC",
            "Variants":                  {
               "Extended": "DISTRICT OF COLUMBIA",
               "ISO": "DC",
               "Abbreviation": "DC"
            },
            "Status":                  {
```

```
                     "MatchStatus": "8",
                     "ResultStatus": "1",
                     "PostalRelevance": "2"
                  }
               },
                              {
                  "Value": "DISTRICT OF COLUMBIA",
                  "Variants": {"Extended": "DISTRICT OF COLUMBIA"},
                  "Status":                  {
                     "MatchStatus": "0",
                     "ResultStatus": "6",
                     "PostalRelevance": "0"
                  }
               }
            ],
            "Country": [            {
               "Code": "USA",
               "Name": "UNITED STATES"
            }]
         },
         "PreformattedData":          {
            "SingleAddressLine": {"Value": "2101 MASSACHUSETTS AVE NW WASHINGTON DC
20008-2811"},
            "PostalDeliveryAddressLines": [{"Value": "2101 MASSACHUSETTS AVE NW"}],
            "PostalFormattedAddressLines":             [
               {"Value": "2101 MASSACHUSETTS AVE NW"},
               {"Value": "WASHINGTON DC 20008-2811"}
            ],
            "PostalLocalityLine": {"Value": "WASHINGTON DC 20008-2811"}
         }
      }]}]
   }]
}
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified United States addresses.

The following table lists the types of enrichments that you can add to a United States address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid United States addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Enrichments	You can retrieve the following enrichments for valid United States addresses: - CBSAID. A five-digit number that identifies a Core-Based Statistical Area (CBSA). A CBSA is an urban area with a population greater than 10,000. - CensusBlockGroup. A 12-digit number that identifies a Census Block Group. A Census Block Group is a group of Census Blocks that share the same first digit. - CensusBlockNo. A four-digit number that identifies a Census Block. A Census Block is the smallest entity for which the Census Bureau collects census information. - CensusCountyFips. A three-digit number that identifies a county in the United States. The United States Census Bureau provides the county information. - CensusCountyName. The name of the county that the Census Bureau provides. - CensusTractNo. A six-digit number that identifies a Census Tract. A Census Tract is a statistical subdivision of a county. - CountyFipsCode. A three-digit number that identifies a county in the United States. The United States Postal Service (USPS) provides the county information. - FinanceNumber. A six-digit number, where the first two digits identify the state and the final four digits identify the post office or postal facility. - MCDID. A five-digit number that identifies a Minor Civil Division (MCD). An MCD is a political or administrative area within a county. - MSAID. A four-digit number that identifies a Metropolitan Statistical Area (MSA). An MSA is an urban area with a population greater than 50,000. - PlaceFipsCode. A five-digit number that identifies a locality within a state. - StateFipsCode. A two-digit number that identifies a state in the United States.
Global Enrichments	You can retrieve the following enrichments for valid United States addresses: - TimeZoneCode. A one- to three-character numeric value that indicates the offset between the address time zone and Greenwich Mean Time. - TimeZoneName. A three-character code that identifies the time zone to which the address belongs.
Geocoding	You can perform geocoding for United States addresses to the following levels of precision: - StreetCenter - ArrivalPoint - Rooftop

Address Verification Tips and Best Practices

Use the following tips and best practices when you verify United States addresses:

- The United States Postal Service recognizes a range of Spanish words and abbreviations.

The following table lists common Spanish address terms and the corresponding abbreviations:

Spanish Words	Abbreviation
AVENIDA	AVE
CALLE	CLL
CAMINITO	CMT
CAMINO	CAM
CERRADA	CER
CIRCULO	CIR
ENTRADA	ENT
PASEO	PSO
PLACITA	PLA
RANCHO	RCH
VEREDA	VER
VISTA	VIS

- If your input data contains sub-building data and street data in a single discrete field, Address Verification can parse the sub-building data from the street data.
- When you enter an address in a partially-fielded input format, limit the amount of non-address data in the postal delivery address line input.

  For example, the following input address includes non-address data:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {}
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": {
          "Locality": [ { "Value": "JOHNSTOWN" } ],
          "PostalCode": [ { "Value": "15905" } ],
          "AdministrativeDivision": [ { "Value": "PA" } ],
          "Country": "USA"
        },
        "PreformattedData": {
          "PostalDeliveryAddressLines": [
            "HILL TOP DENTAL", "1X CREDIT CARD PURCHASE", "1736 LYTER DRIVE"
          ]
        }
      }
    ]
  }
}
```

  In the `PostalDeliveryAddressLines` elements, the detail `1X CREDIT CARD PURCHASE` indicates non-address data.

- Do not include locality, postal code, province, or country information in postal delivery address line input. For best results, enter the information in postal locality line input or in discrete input fields.
- To verify an address to delivery point level, submit the address in certified mode. Address Verification can return Delivery Point Validation (DPV) fields for valid addresses in certified mode.
- To determine if a physical access point exists for an address, submit the address in certified mode and select the Door Not Accessible (DNA) element. A DNA address might not provide a door for mail delivery, or the mailbox might reside behind a locked gate.
- To verify or update an address to the form that the 911 emergency system recognizes, submit the address in certified mode.

  Use certified mode to convert rural route or highway addresses to street addresses.
- For best results, provide a descriptor when you enter sub-building information.

  For example, enter the sub-building information as `100 Main St Apt 10` or `100 Main St #10` instead of `100-10 Main St` or `100 Main St 10`.
- For best results, use the ISO two-letter code or three-letter code to identify the country.

Frequently Asked Questions

Review the following frequently asked questions about address verification for United States data:

How does Address Verification handle ZIP+4 Codes in United States addresses in QuickCapture mode?

Address Verification returns only the five-digit post code in the address suggestions, and omits the additional four digits when you verify an address that contains a ZIP+4 Code in QuickCapture mode.

Use batch or certified mode to receive ZIP+4 Codes in the address output when you verify an address that contains a ZIP+4 Code.

Should I include county information in an input address?

No, a United States address does not require county information in the input.

How does Address Verification handle state and county names in United States addresses?

Address Verification can return the state and county names as an array in the AdministrativeDivision address element.

The first AdministrativeDivision array value indicates the state name. The second AdministrativeDivision array value indicates the county name. Address Verification does not use the third AdministrativeDivision array value for United States addresses.

Note: The county name does not form part of a regular United States postal address.

For example, the following address is valid and does not contain the county name:

```
502 FAIRVIEW ST
ABBEVILLE LA 70510
USA
```

The following sample code shows the state and county names that Address Verification can return for the address:

```
"AdministrativeDivision": [
    {
        "Value": "LA",
        "Variants": {
            "Extended": "LOUSIANA",
            "ISO": "LA",
            "Abbreviation": "LA"
        },
```

```
            "Status": {
                "MatchStatus": "0",
                "ResultStatus": "6",
                "PostalRelevance": "2"
                }
        },
        {
            "Value": "VERMILION",
            "Variants": { "Extended": "VERMILION" },
            "Status": {
                "MatchStatus": "0",
                "ResultStatus": "6",
                "PostalRelevance": "0"
                }
        }
    ],
```

How do I get the CensusCountyFips and CensusCountyName codes for a United States address?

To receive the CensusCountyFips and CensusCountyName codes for a United States address, install the latest supplementary reference data files for United States and ensure that valid license keys are available for the data files.

For more information about the codes, see "Address Enrichments" on page 8.

How do I get the CountyFipsCode code for a United States address?

To receive the CountyFipsCode code for a United States address, install the latest supplementary reference data files for United States and ensure that valid license keys are available for the data files.

For more information about the code, see "Address Enrichments" on page 8.

Which agency does Informatica receive census data from?

Informatica receives the census data from the United States Census Bureau.

Which version or year of the census data does Informatica use in its reference data?

Currently, Informatica uses the 2020 census data.

Can Address Verification tell whether an input address is a business or a home address?

The Residential Data Indicator indicates whether the address points to a residence or a business.

Does Address Verification reference data include DPV data?

Address Verification reference data for the United States contains DPV data. The certified DPV database has information that helps you check whether a ZIP+4 coded address is in the USPS delivery file as a known address record.

Does Address Verification return CASS certification codes for United States addresses?

Yes. When you verify an address in certified mode, Address Verification returns the CASS certification codes for United States addresses. For information about CASS certification codes, see *Address Verification (On-Premises) Developer Guide*.

Reference Address Databases

Informatica updates United States reference databases at regular intervals. For example, Informatica updates batch, interactive, and certified databases on a monthly basis. You can download the latest database packages from the Informatica Database Download portal at the following address: https://dqc.addressdoctor.com/SSO/Default. To enable your account on the portal, contact your Informatica account representative.

For news about reference address databases and software updates for Address Verification, subscribe to the Informatica DaaS Newsletter from Informatica.

Informatica Address Verification provides the following reference databases for United States addresses:

- `USA_ADV_VRF_BIA_000_6_x_0.MD6`. Batch and interactive data.
- `USA_ADV_VRF_QCP_000_6_x_0.MD6`. QuickCapture data.
- `USA_ADV_VRF_C01_000_6_x_0.MD6` through `USA_ADV_VRF_C07_000_6_x_0.MD6`. Certified data.
- `USA_ADV_RLK_GTA_000_6_x_0.MD6`. GeocodeToAddress data.
- `USA_ADV_ENR_GST_000_6_x_0.MD6`. StreetCenter geocoding data.
- `USA_ADV_ENR_GAP_000_6_x_0.MD6`. ArrivalPoint geocoding data.
- `USA_ADV_ENR_GRT_000_6_x_0.MD6`. Rooftop geocoding data.
- `USA_ADV_ENR_EN1_000_6_x_0.MD6`. Supplementary data.
- `USA_ADV_ENR_CAM_000_6_x_0.MD6`. CAMEO data.

The value `x` in the `MinVersion` variable of the database file names indicates the minor version of the release which can contain one of the following values: `1`, `2`, or `3`

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification uses a cloud interface.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, check the following help modules on the Informatica Documentation Portal:

- Address Verification (On-Premises) 6.x
- Address Verification (Cloud) 6.x

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